Change in the 17th century was slow at first. Despite the efforts of Dutch ruler Maurice of Nassau (1567–1625) and Swedish king Gustav Adolf (1594–1632) to improve army organization, it was only through the second half of the century that regular armed forces, permanently employed by the state and fitted with standard uniforms, gradually became the norm. Mercenary captains and aristocratic cavalrymen were transmuted into officers in the standing army, and received a defined rank. Mercenaries continued to be employed only in the form of companies of foreign soldiers hired from their ruler, rather than from private entrepreneurs. Perfected in the Prussian army of Frederick the Great (1713–86), draconian discipline and rigorous drill sought to make the infantryman, recruited from the lowest levels of society, into an automaton capable of holding firm on the battlefield regardless of his personal qualities. More regular pay and supply reduced the plunder and looting previously associated with military operations.

INFANTRY TACTICS

From the start of the 16th century and well into the second half of the 17th century, infantry tactics in Europe were based upon the use of tight squares of pikemen flanked by soldiers with firearms. At first, the firearms were used as an ancillary to the all-important pikes, but over time the number and effectiveness of musketeers grew, while the significance of the pikemen diminished. In the last decades of the 17th century the introduction of the flintlock musket and of socket bayonets led to the disappearance of the pike from the battlefield. By the early 18th century, European infantry were formidable, disciplined bodies, trained to fire in volleys and march unarmoured into cannon and musket fire.

VARIED CAVALRY

The cavalry, seen as more prestigious, remained the standard place for the aristocracy, even if the increasing size of European armies meant that its lower ranks had to be opened to commoners. Horsemen struggled to find the most effective battlefield tactics, however, in the face of increasing firepower. The full armour and lance of the medieval knight had been abandoned by the end of the 16th century, but heavy cavalry still remained decisive shock troops – Gustav Adolf, the king of Sweden, was killed leading a cavalry charge at Lutzen in 1632. Along with a sword, cavalrymen adopted firearms such as pistols or carbines – the famous elite French musketeers were mounted troops.

Armoured dragoons, riding to battle but dismounting in order to fight with their firearms, became an important element of European armies. So too did light cavalry, used for reconnaissance, skirmishing, and raiding. The example for these dashing horsemen was set by the impressive Polish winged hussars and the Russian Cossacks.

Mughal battle
The forces of Mughal emperor Akbar (reigned 1556–1605) fought with curved, slashing *talwar* swords in the warrior spirit of Islam, but combined these traditional weapons with up-to-date artillery.



OUTSIDE EUROPE

The surprisingly easy victories of the Spanish conquistadors over the large armies of Central America and Peru in the 16th century could give a false impression of European military superiority over the rest of the world. In fact, for much of this period there were armies outside Europe equal or superior in their organization, technology, and tactics. In the 1520s, when Hernán Cortés and his followers were crushing the Aztec Empire in Mexico, Muslim Ottoman forces shattered a Christian Hungarian army at Mohács and laid siege to Vienna, the capital of the Habsburg Empire.

Gunpowder weapons were successfully adopted in both Africa and Asia. Moroccan forces used muskets and cannon on military expeditions south of the Sahara in the 1590s. The Manchu armies that conquered China in the mid-17th century deployed large cannon to decisive effect. In Japan, arquebusiers firing in volleys gave Oda Nobunaga victory at the battle of Nagashino in 1575. The creation of the Mughal Empire in India is partly attributable to skilful employment of cannon by the empire's founder, Babur, at the battle of Panipat in 1526.

The major Islamic states of the Mughals, the Ottomans, and Safavid Persia at best effectively combined the virtues of a traditional warrior ethic with up-to-date weaponry. The Ottoman army had skirmishing light horsemen fighting in a long-established Central Asian style, heavy cavalry based on a semi-feudal system, high-quality disciplined infantry (the janissary slave-soldiers), and varied artillery. The decay of the once excellent Ottoman forces, well under way by the 1650s, reflected not so much specifically military failings as an irreversible institutional decay within the Ottoman Empire.

By the end of the 17th century there were already clear signs that European armies were gaining a lead over their extra-European rivals. Armies of the Muslim world were slow to replace the matchlock musket with the flintlock, while the Chinese, the original inventors of gunpowder weapons, came to depend upon European experts to maintain their artillery. Asian armies began to look unwieldy and disorganized compared to the disciplined, uniformed standing armies of 18th-century Europe – formidable forces on an open battlefield. Discipline and firepower opened the way for the age of European world domination.

